

Supply chain performance analysis

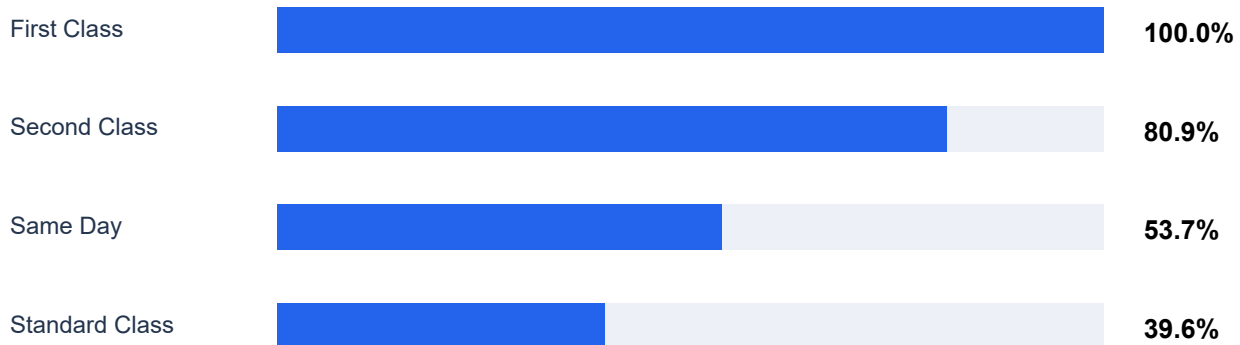
AMINE BELARBI - ENCG BUSINESS ADMINISTRATION

Which shipping modes should operations investigate first?

An educational case study connecting an existing Excel dashboard, reproducible SQL and Power BI project definitions. Historical sample; no employer confidential data.

15,000	9,656	9,209	52.2%
Order lines	Distinct orders	Eligible orders	Late rate

Rate and volume imply different priorities



First Class has the highest observed late rate (100.0%). Standard Class accounts for the most late orders (2,657; 55.3% of eligible late orders).

Investigate schedule/service-promise assumptions in high-rate modes, then review the high-volume source of lateness. The data does not identify carrier root causes or demonstrate savings.

Sample period: 2015-01-01 to 2018-01-31. Unknown selection method; partial orders possible. Source: DataCo / Mendeley Data, CC BY 4.0.

Method, tools and verification

1. Check the grain before calculating KPIs

The 15,000 unique order items group into 9,656 orders. Validate that dates, status, mode, market, region and shipping durations agree within each order before deduplication.

2. Use an explicit eligible population

Exclude 447 canceled/suspected-fraud orders from the shipping-rate denominator. Keep them in the fact table for reconciliation. Late = actual shipping days greater than scheduled shipping days.

3. Connect Excel, SQL and the BI model

Excel: original pivot/chart workbook retained in a redacted public copy. Corrected workbook calculates order-level metrics with COUNTIFS, AVERAGEIFS and linked charts.

SQL: typed line table, order view, quality checks, CTEs, LAG and DENSE_RANK. Aggregate financial ratios use sums rather than averages of row ratios.

Power BI: order and line facts share date, shipping-mode and market dimensions. Twelve DAX measures and three PBIR report pages are authored. Native Desktop refresh and rendering remain pending.

Shipping-mode comparison

Mode	Eligible	Late	Late rate	Avg late days
First Class	892	892	100.0%	1.00
Second Class	1,459	1,181	80.9%	2.51
Same Day	147	79	53.7%	1.00
Standard Class	6,711	2,657	39.6%	1.50

Validation and honest limits

SQL totals reconcile with an independent Python calculation. Duplicate item keys, inconsistent order attributes, late-status contradictions and reversed dates are checked. Source rows are order lines, not confirmed physical shipments. Shipping duration is not a verified customer receipt timestamp. OTIF and inventory turnover cannot be calculated from the available inputs.

GitHub: project files, methodology and practice guide

Source: Constante, Silva and Pereira (2019), DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS, Mendeley Data V5, DOI 10.17632/8gx2fvg2k6.5.

Original Excel work by Amine. Extended SQL, model/report definitions, corrected workbook and documentation developed with AI assistance. No claimed measured operational improvement.